

A candidate proof of Conjecture 1.2 in Isoperimetry, Stability, and Irredundance in Direct Products

github.com/graph-theory-AI — machine-generated candidate writeup

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Abstract

A tensor-rank bound retaining the lonely vertices, combined with rank-nonincreasing coordinate compression, proves the conjecture for all complete multipartite factors. This candidate proof was generated by GPT-6 Astra in a single model pass for the Graph Theory AI project. It was selected from the campaign because the model marked it `would_publish: true`; it has not been checked by the adversarial referee, a human mathematician, or a novelty search.

1 The problem and the claimed result

This document concerns [Conjecture 1.2](#) from *Isoperimetry, Stability, and Irredundance in Direct Products*.

1.1 Problem statement

Conjecture. If $H_1, \dots, H_n$ are complete multipartite graphs and $G = \prod_{i=1}^n H_i$, then $\alpha(G) = IR(G)$.

1.2 Catalog context

The authors propose this as a further strengthening of Burcroff’s Conjecture 1.1, removing the assumption that the complete multipartite graphs in the direct product are balanced. It is stated at the end of the introduction after the main results are summarised.

1.3 Claimed result

Model verdict: proved (confidence: high).

A tensor-rank bound retaining the lonely vertices, combined with rank-nonincreasing coordinate compression, proves the conjecture for all complete multipartite factors.

The model’s own caveat: Literature priority is unchecked; the proof below is self-contained.

2 Model writeup

The text below is the model output verbatim, apart from moving its machine-readable verdict block into the summary above and shifting Markdown heading levels for this document. Mechanical TeX defects and equation tags were normalized where needed for compilation.

2.1 Theorem

Let $H_1, \dots, H_n$ be finite complete multipartite graphs, where $n \geq 1$, and let

$$G = \prod_{i=1}^n H_i$$

be their direct product. Then

$$\alpha(G) = IR(G).$$

The tensor argument in the previous attempt can be strengthened substantially: if an irredundant set has t social vertices and its lonely vertices have tensor rank r , then

$$t + 2r \leq 2^n.$$

The second ingredient below is a compression argument showing that a nonempty independent family of rank r falls short of maximum weight by at least $2^n - 2r$. These two inequalities fit together exactly.

All ingredients used from the previous attempt are proved afresh. In particular, the proof does not rely on its blocker inequality or its low-dimensional classifications.

2.1.1 1. Irredundance and the bipartite reduction

Write $N[v]$ for the closed neighborhood of v . A set $S \subseteq V(G)$ is **irredundant** if, for every $v \in S$, there exists a vertex p such that

$$N[p] \cap S = \{v\}.$$

Every independent set is irredundant, so it suffices to prove

$$|S| \leq \alpha(G)$$

for every irredundant set S .

Call a vertex of S **lonely** if it has no neighbor in S , and **social** otherwise. A social vertex v must have an external private neighbor

$$p_v \in V(G) \setminus S, \quad N(p_v) \cap S = \{v\}.$$

Private neighbors chosen for distinct social vertices are distinct.

We first record an elementary reduction.

2.1.1.1 Lemma 1 Every finite bipartite graph satisfies $IR(G) = \alpha(G)$.

2.1.1.1.1 Proof Let $X \sqcup Y$ be a bipartition, and let S be irredundant. Let L_Y be the lonely vertices in $S \cap Y$. For each social $y \in S \cap Y$, choose an external private neighbor $p_y \in X \setminus S$.

Then

$$I = (S \cap X) \cup L_Y \cup \{p_y : y \in S \cap Y \text{ is social}\}$$

is independent. Indeed, the vertices outside L_Y all lie in X ; lonely vertices have no neighbors in S ; and privacy ensures that every p_y is nonadjacent to L_Y . The chosen private neighbors are distinct and outside S , so $|I| = |S|$. Thus $|S| \leq \alpha(G)$. $\square$

If a factor has no vertices, the theorem is immediate. If a factor has one nonempty part, the product is edgeless. If a factor has two nonempty parts, the product is bipartite.

Henceforth, assume that every factor has at least three nonempty parts.

2.1.2 2. The weighted quotient and its tensor representation

Write the parts of H_i as

$$P_{i,1}, \dots, P_{i,q_i}, \quad s_{i,a} = |P_{i,a}|,$$

where $q_i \geq 3$. Order the parts so that

$$s_{i,1} \geq s_{i,2} \geq \dots \geq s_{i,q_i} \geq 1.$$

Set

$$\Omega = \prod_{i=1}^n [q_i].$$

For $x = (x_1, \dots, x_n) \in \Omega$, define the cell and its weight by

$$C_x = \prod_{i=1}^n P_{i,x_i}, \quad w(x) = |C_x| = \prod_{i=1}^n s_{i,x_i}.$$

Each cell is independent. Two cells are completely joined exactly when their labels differ in every coordinate.

Consequently, the quotient graph is

$$Q = \prod_{i=1}^n K_{q_i},$$

and

$$\alpha(G) = \alpha_w(Q) := \max_{\mathcal{J} \text{ independent in } Q} W(\mathcal{J}), \quad W(\mathcal{J}) = \sum_{x \in \mathcal{J}} w(x). \quad (2.1)$$

Indeed, an independent set can be enlarged to contain every vertex of each cell it meets.

For each coordinate and label, put

$$u_{i,a} = (1, a) \in \mathbb{R}^2,$$

and define

$$z_x = \bigotimes_{i=1}^n u_{i,x_i} \in V := (\mathbb{R}^2)^{\otimes n}.$$

For $\mathcal{A} \subseteq \Omega$, write

$$M(\mathcal{A}) = \text{span}\{z_x : x \in \mathcal{A}\}, \quad \rho(\mathcal{A}) = \dim M(\mathcal{A}).$$

Equip V with the tensor product of the determinant forms:

$$\beta\left(\bigotimes_i v_i, \bigotimes_i v'_i\right) = \prod_{i=1}^n \det(v_i, v'_i).$$

This bilinear form is nondegenerate: its matrix is a tensor power of the invertible 2×2 determinant-form matrix. Moreover,

$$\beta(z_x, z_y) = \prod_{i=1}^n (y_i - x_i),$$

so

$$\beta(z_x, z_y) \neq 0 \iff x \text{ and } y \text{ are adjacent in } Q. \quad (2.2)$$

2.1.3 3. The rank-defect bound for social vertices

2.1.3.1 Lemma 2 Let $S \subseteq V(G)$ be irredundant. Let L be its lonely vertices, let $t = |S \setminus L|$, and let

$$\mathcal{A} = \{x \in \Omega : C_x \cap L \neq \emptyset\}.$$

Then

$$\boxed{t + 2\rho(\mathcal{A}) \leq 2^n.} \quad (3.1)$$

2.1.3.1.1 Proof Enumerate the social vertices as $v_1, \dots, v_t$, with labels $x_1, \dots, x_t$, and choose external private neighbors $p_1, \dots, p_t$, with labels $y_1, \dots, y_t$.

Privacy and (2.2) give

$$\beta(z_{y_j}, z_{x_k}) = \begin{cases} \neq 0, & j = k, \\ 0, & j \neq k. \end{cases} \quad (3.2)$$

Every p_j is nonadjacent to every lonely vertex, so

$$\beta(z_{y_j}, M(\mathcal{A})) = 0. \quad (3.3)$$

Thus the vectors $z_{x_1}, \dots, z_{x_t}$ are linearly independent modulo $M(\mathcal{A})$: pairing a relation

$$\sum_{k=1}^t c_k z_{x_k} \in M(\mathcal{A})$$

with z_{y_j} forces $c_j = 0$. Therefore

$$\dim(M(\mathcal{A}) + \text{span}\{z_{x_1}, \dots, z_{x_t}\}) = \rho(\mathcal{A}) + t. \quad (3.4)$$

The lonely vertices form an independent set, and every social vertex is nonadjacent to every lonely vertex. Hence the space in (3.4) lies in

$$M(\mathcal{A})^\perp = \{v \in V : \beta(v, m) = 0 \text{ for all } m \in M(\mathcal{A})\}.$$

Nondegeneracy gives

$$\dim M(\mathcal{A})^\perp = 2^n - \rho(\mathcal{A}).$$

Combining this with (3.4) proves (3.1). $\square$

In particular, if $L = \emptyset$, then

$$|S| \leq 2^n. \quad (3.5)$$

2.1.4 4. Coordinate compression does not increase tensor rank

For a coordinate i , define the compression $C_i \mathcal{A}$ as follows. Fix all coordinates except i . If the resulting fiber contains k members of $\mathcal{A}$, replace their i -th coordinates by

$$1, 2, \dots, k.$$

2.1.4.1 Lemma 3 For every $\mathcal{A} \subseteq \Omega$:

1. If $\mathcal{A}$ is independent, then $C_i \mathcal{A}$ is independent.
2. $W(C_i \mathcal{A}) \geq W(\mathcal{A})$.
3. $\rho(C_i \mathcal{A}) \leq \rho(\mathcal{A})$.

2.1.4.1.1 Proof Independence. Suppose two rest-coordinate tuples differ in every coordinate. If their two original fibers are both nonempty, independence requires every label in one fiber to equal every label in the other. Thus both fibers must be the same singleton. Compression replaces both singleton labels by 1, so it cannot create an adjacent pair between these fibers. This proves the first assertion.

Weight. In a fixed fiber, the weight contributed by coordinate i changes from a sum of k part sizes to

$$s_{i,1} + \dots + s_{i,k},$$

which is at least as large. The remaining coordinates contribute a fixed positive factor.

Rank. Identify V with $\mathbb{R}^2 \otimes V_{-i}$. For each rest-coordinate tuple x , let $z_x \in V_{-i}$ be its tensor vector, and let $A_x \subseteq [q_i]$ be the original fiber.

Put

$$T = \text{span}\{z_x : A_x \neq \emptyset\}, \quad E = \text{span}\{z_x : |A_x| \geq 2\}.$$

Let $M = M(\mathcal{A})$, and let

$$\pi = \ell \otimes \text{id}, \quad \ell(a, b) = a.$$

Because $\ell(u_{i,a}) = 1$ for every label,

$$\pi(M) = T.$$

If $|A_x| \geq 2$, subtracting the tensors corresponding to two labels in that fiber shows that

$$(0, 1) \otimes z_x \in M.$$

It follows that

$$(0, 1) \otimes E \subseteq M \cap \ker \pi.$$

Rank-nullity therefore gives

$$\rho(\mathcal{A}) \geq \dim T + \dim E. \quad (4.1)$$

After compression, a nonempty singleton fiber uses $u_{i,1}$, while a fiber of size at least two spans $\mathbb{R}^2$ in coordinate i . Consequently,

$$M(C_i \mathcal{A}) = u_{i,1} \otimes T \oplus (0, 1) \otimes E,$$

and hence

$$\rho(C_i \mathcal{A}) = \dim T + \dim E.$$

Together with (4.1), this proves the rank assertion. $\square$

Repeatedly compress any coordinate whose compression changes the family. This process terminates, since every change strictly decreases the positive integer

$$\sum_{x \in \mathcal{A}} \sum_{i=1}^n x_i.$$

The resulting family $\mathcal{D}$ is a down-set in the coordinatewise order. If $\mathcal{A}$ was independent, then

$$\mathcal{D} \text{ is independent,} \quad W(\mathcal{D}) \geq W(\mathcal{A}), \quad \rho(\mathcal{D}) \leq \rho(\mathcal{A}). \quad (4.2)$$

2.1.5 5. The weighted rank-defect inequality

This is the second main ingredient.

2.1.5.1 Lemma 4 Suppose $q_i \geq 3$ for every i . For every nonempty independent family $\mathcal{A} \subseteq \Omega$,

$$\boxed{\alpha_w(Q) - W(\mathcal{A}) \geq 2^n - 2\rho(\mathcal{A}).} \quad (5.1)$$

2.1.5.1.1 Proof Compress $\mathcal{A}$ to a down-set $\mathcal{D}$ as in (4.2).

The tensors

$$\{z_b : b \in \{1, 2\}^n\}$$

form a basis of V . Because $\mathcal{D}$ is a down-set, every $z_x, x \in \mathcal{D}$, is a linear combination of basis tensors whose labels also lie in $\mathcal{D}$: expand each u_{i,x_i} in the basis $u_{i,1}, u_{i,2}$, noting that every label appearing in the expansion is coordinatewise at most x . Therefore

$$\rho(\mathcal{D}) = |\mathcal{D} \cap \{1, 2\}^n|. \quad (5.2)$$

For $T \subseteq [n]$, define the binary tuple

$$b(T)_i = \begin{cases} 1, & i \in T, \\ 2, & i \notin T, \end{cases}$$

and put

$$\mathcal{F} = \{T \subseteq [n] : b(T) \in \mathcal{D}\}.$$

Then $\mathcal{F}$ is upward-closed and

$$|\mathcal{F}| = \rho(\mathcal{D}). \quad (5.3)$$

Since $\mathcal{D} \neq \emptyset$, it contains $(1, \dots, 1)$, so

$$[n] \in \mathcal{F}. \quad (5.4)$$

Moreover, $\mathcal{F}$ is intersecting. Indeed, if $T, U \in \mathcal{F}$ were disjoint, then $T \subseteq [n] \setminus U$, so upward closure would give

$$[n] \setminus U \in \mathcal{F}.$$

But $b(U)$ and $b([n] \setminus U)$ differ in every coordinate, contradicting independence of $\mathcal{D}$.

For any intersecting family $\mathcal{H} \subseteq 2^{[n]}$, define

$$E(\mathcal{H}) = \{x \in \Omega : \{i : x_i = 1\} \in \mathcal{H}\}.$$

This is independent: any two of its members have a common coordinate equal to 1.

Also,

$$\mathcal{D} \subseteq E(\mathcal{F}). \quad (5.5)$$

To see this, take $x \in \mathcal{D}$ and let $T = \{i : x_i = 1\}$. Then $b(T) \leq x$ coordinatewise, so $b(T) \in \mathcal{D}$.

Extend $\mathcal{F}$ to an inclusion-maximal intersecting family $\mathcal{F}^*$ in $2^{[n]}$. We need the elementary fact

$$|\mathcal{F}^*| = 2^{n-1}. \quad (5.6)$$

Indeed, a maximal intersecting family is upward-closed. If $T \notin \mathcal{F}^*$, maximality supplies some $U \in \mathcal{F}^*$ disjoint from T . Hence $U \subseteq [n] \setminus T$, and upward closure gives $[n] \setminus T \in \mathcal{F}^*$. An intersecting family cannot contain both members of a complementary pair, proving (5.6).

The total weight of all tuples whose coordinates equal 1 precisely on T is

$$m(T) = \prod_{i \in T} s_{i,1} \prod_{i \notin T} \left(\sum_{a=2}^{q_i} s_{i,a} \right). \quad (5.7)$$

Every $T \in \mathcal{F}^* \setminus \mathcal{F}$ differs from $[n]$, by (5.4). Thus $i \notin T$ for some coordinate i . Since $q_i \geq 3$ and all part sizes are positive integers,

$$m(T) \geq 2. \quad (5.8)$$

The pattern classes in (5.7) are disjoint, so

$$\begin{aligned} W(E(\mathcal{F}^*)) - W(E(\mathcal{F})) &= \sum_{T \in \mathcal{F}^* \setminus \mathcal{F}} m(T) \\ &\geq 2(2^{n-1} - |\mathcal{F}|) \\ &= 2^n - 2\rho(\mathcal{D}). \end{aligned} \quad (5.9)$$

Finally, $E(\mathcal{F}^*)$ is independent, and

$$W(E(\mathcal{F})) \geq W(\mathcal{D}) \geq W(\mathcal{A}).$$

Consequently,

$$\begin{aligned} \alpha_w(Q) - W(\mathcal{A}) &\geq W(E(\mathcal{F}^*)) - W(E(\mathcal{F})) \\ &\geq 2^n - 2\rho(\mathcal{D}) \\ &\geq 2^n - 2\rho(\mathcal{A}), \end{aligned}$$

as required. $\square$

2.1.6 6. Irredundant sets with a lonely vertex

Let S be irredundant, with lonely set $L \neq \emptyset$, social count t , and lonely-label support $\mathcal{A}$.

We have $|L| \leq W(\mathcal{A})$. Combining Lemmas 2 and 4 gives

$$\begin{aligned} |S| &= |L| + t \\ &\leq W(\mathcal{A}) + 2^n - 2\rho(\mathcal{A}) \\ &\leq \alpha_w(Q) \\ &= \alpha(G). \end{aligned} \quad (6.1)$$

Thus every remaining case has no lonely vertices.

2.1.7 7. Irredundant sets with no lonely vertices

If S has no lonely vertices, Lemma 2 gives

$$|S| \leq 2^n. \quad (7.1)$$

2.1.7.1 7.1 At least three factors For $n \geq 3$, fixing one coordinate to lie in a particular part gives an independent set of size at least 3^{n-1} . Hence

$$\alpha(G) \geq 3^{n-1} \geq 2^n \geq |S|. \quad (7.2)$$

2.1.7.2 7.2 One factor For $n = 1$, (7.1) gives $|S| \leq 2$. If some part has size at least two, then $\alpha(G) \geq 2$, and we are done.

Otherwise G is complete. No set of at least two vertices in a complete graph is irredundant: none of its vertices has an external private neighbor. Thus this case is also settled.

2.1.7.3 7.3 Two factors Here $|S| \leq 4$. A coordinate fiber has size at least the order of the other factor, so

$$\alpha(G) \geq \max\{|V(H_1)|, |V(H_2)|\}.$$

Since each factor has at least three parts, this is at least four unless

$$H_1 = H_2 = K_3.$$

It remains to rule out a four-vertex all-social irredundant set in $K_3 \times K_3$.

Identify its vertices with the cells of a 3×3 array. Adjacency means differing in both row and column. A private neighbor (r', c') of a selected cell (r, c) must lie outside S , satisfy $r' \neq r$, $c' \neq c$, and have its row and column cover the other three selected cells.

No row can contain three selected cells: a private neighbor of one of them lies in another row and cannot avoid adjacency to the other two, whose columns are distinct. Thus the row-count pattern of a four-cell set is either $(2, 2, 0)$ or $(2, 1, 1)$.

- **Pattern $(2, 2, 0)$.**

Suppose one occupied row i contains columns a, b , and the other occupied row is j . A private neighbor of (i, a) is forced to be (j, b) , and a private neighbor of (i, b) is forced to be (j, a) . Both must be unselected. Thus the two selected cells in row j must avoid both columns a, b , impossible with only three columns.

- **Pattern $(2, 1, 1)$.**

Suppose row i contains $(i, a), (i, b)$, and the other selected cells are $(j, c), (k, d)$, with i, j, k distinct. A private neighbor of (i, a) has column b , so at least one of c, d equals b . Similarly, at least one equals a . Therefore

$$\{c, d\} = \{a, b\}.$$

Relabel so that the remaining cells are $(j, a), (k, b)$. A private neighbor of (j, a) , to avoid both selected cells in row i , must have row i ; to avoid (k, b) , it must then have column b . It is therefore the already selected cell (i, b) , a contradiction.

So $|S| \leq 3$ in $K_3 \times K_3$, which has an independent row of size three.

2.1.8 8. Conclusion

Every irredundant set in the stated product has size at most $\alpha(G)$. Since every independent set is irredundant,

$$IR(G) \leq \alpha(G) \leq IR(G).$$

Therefore

$$IR\left(\prod_{i=1}^n H_i\right) = \alpha\left(\prod_{i=1}^n H_i\right)$$

for arbitrary finite complete multipartite factors.

All graph cases are covered: empty or edgeless products, products with a bipartite factor, products with a lonely vertex in the chosen irredundant set, and lonely-free sets in every dimension. No computational check, balanced-case theorem, or additional conjecture is required.

A Provenance and review status

UNREFEREED MODEL OUTPUT. This document was selected solely because GPT-6 Astra labelled its own result `would_publish: true` and returned `proved` or `disproved`. It has not passed the adversarial LLM referee used for the earlier Sol campaign, has not been checked by a human mathematician, and has not been checked for novelty. Treat every mathematical and bibliographic claim as unverified.

Catalog id	1904.02595__00
Catalog entry	Conjecture 1.2
Source corpus	arxiv
Campaign leg	selected second attempts (<code>attacks_retry</code>)
Source paper / entry	Isoperimetry, Stability, and Irredundance in Direct Products
Model verdict	proved (confidence: high)
Selection criterion	<code>would_publish: true</code> and verdict <code>proved</code> or <code>disproved</code>
Model	<code>gpt-6-astra</code> , reasoning effort <code>max</code> , mode= <code>pro</code> , flex service tier
Original artifact	<code>attacks_retry/1904.02595__00/output.md</code>
Independent review	None
arXiv	arXiv:1904.02595
Previous attempt	<code>attacks/1904.02595__00</code> : <code>gpt-5.6-sol</code> verdict <code>partial</code>
Model's caveats	Literature priority is unchecked; the proof below is self-contained.

The generated Markdown and LaTeX sources for this PDF are stored under `to_review_astra/src/1904.02595__00/`.